

Computer Science & IT

Theory of Computation



Regular Languages ✓

Lecture No. 5



By- Ravindra Sir

Recap of Previous Lecture



Topic

Reg Languages

Topic

Topic

Topics to be Covered



Topic

Reg languages

Topic

Topic



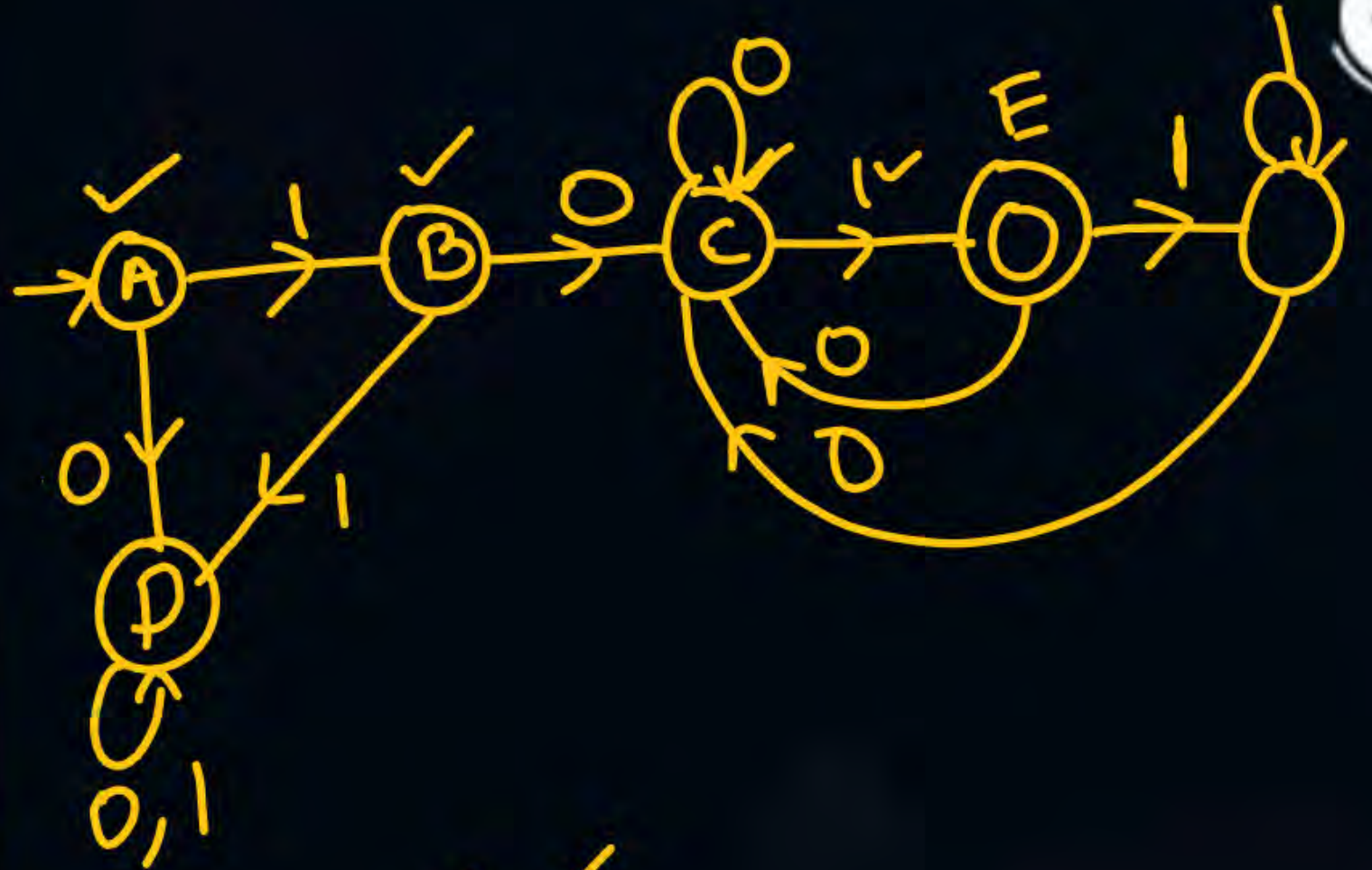
Starting with $\underline{10}$ ending with $\underline{01}$
 $\underline{101 + 10(0+1)^*01}$



$C \rightarrow \underline{10} \underline{1}$
 $\underline{10} \underline{0} \underline{1}$
 $\underline{10} \underline{000} \underline{1}$

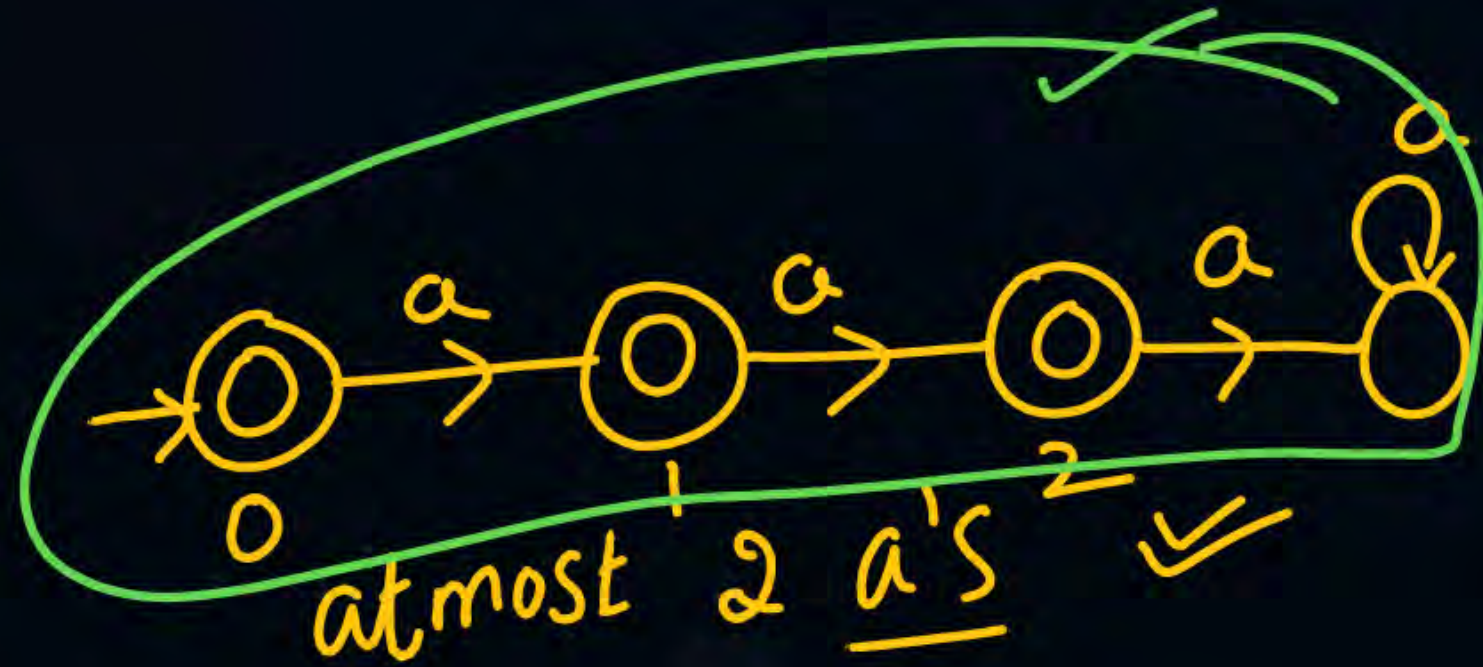
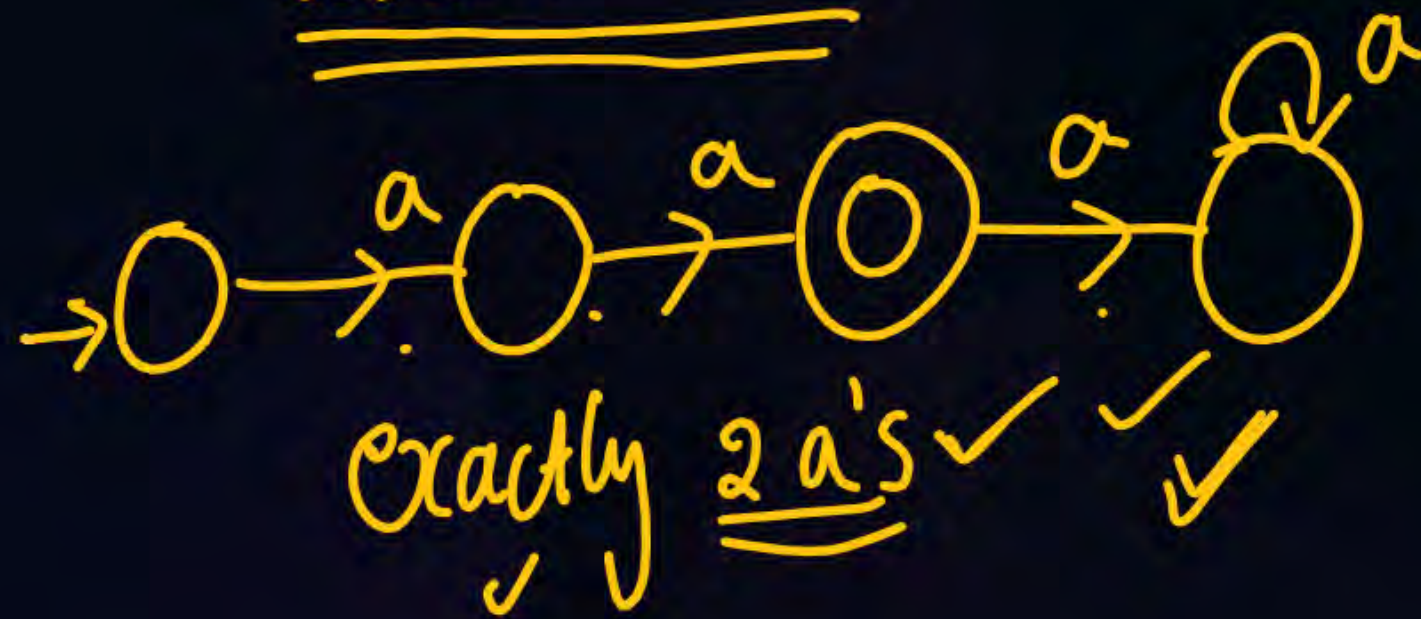
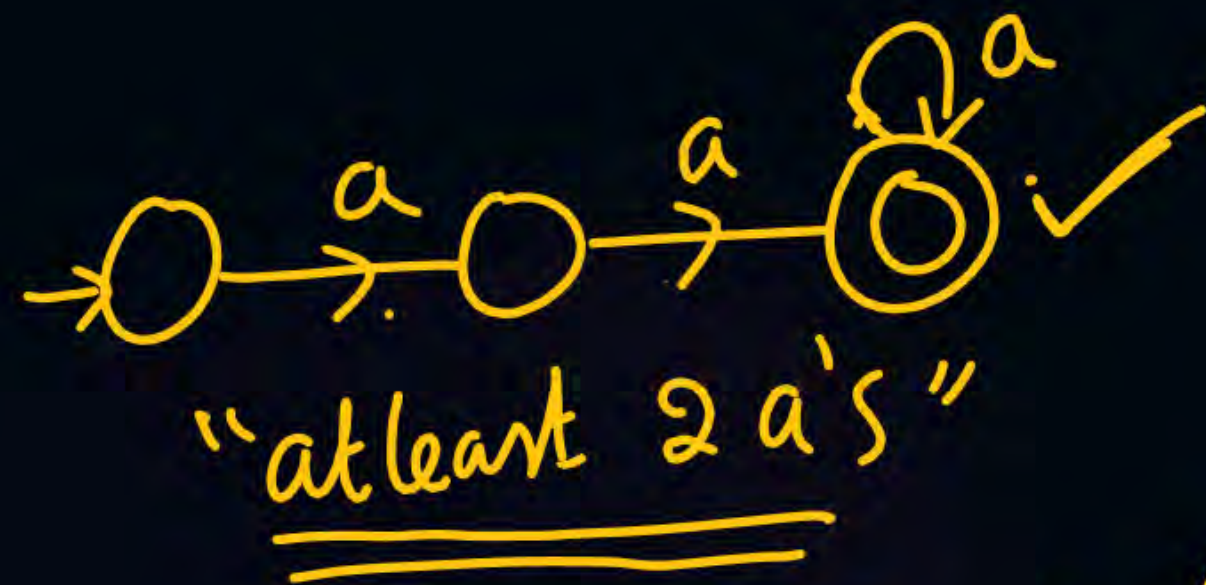


Starting with $\overset{\checkmark}{(10)}$ ending with $\overset{\checkmark}{(01)}$
 $\underline{101 + 10(0+1)^*01}$

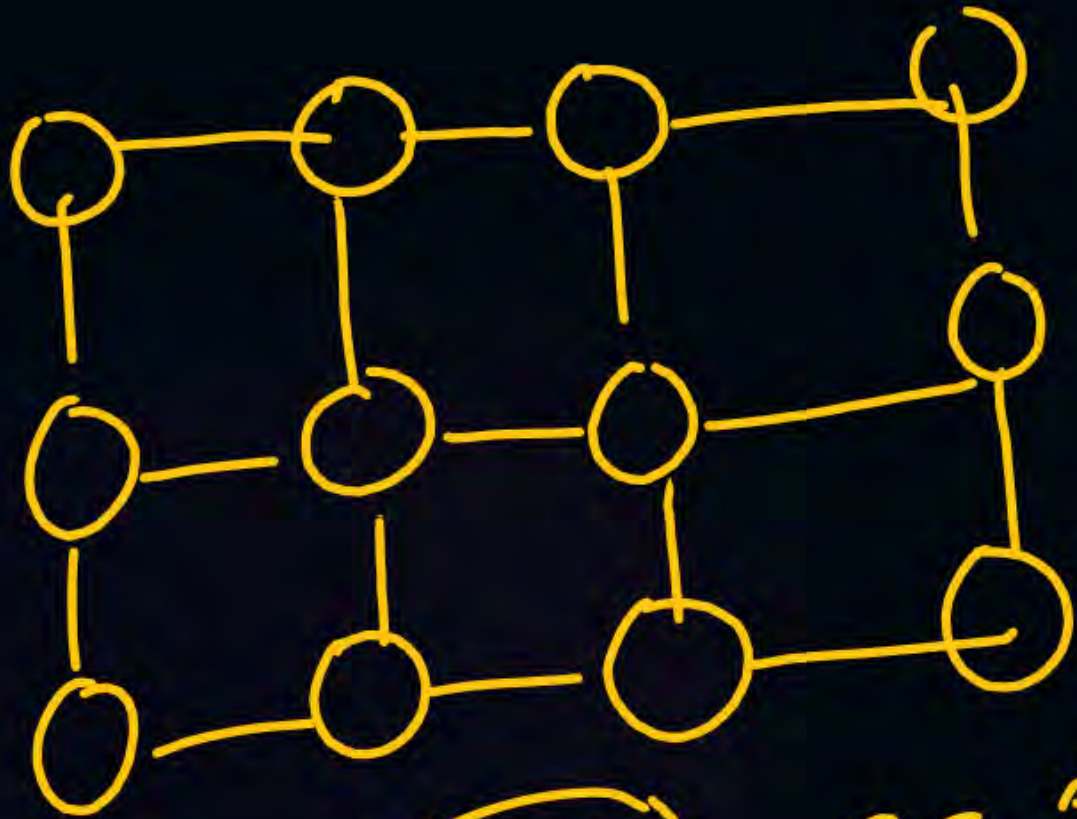


$E \rightarrow 101 \overset{\checkmark}{(111)} \overset{\checkmark}{(01)}$

Grid machines



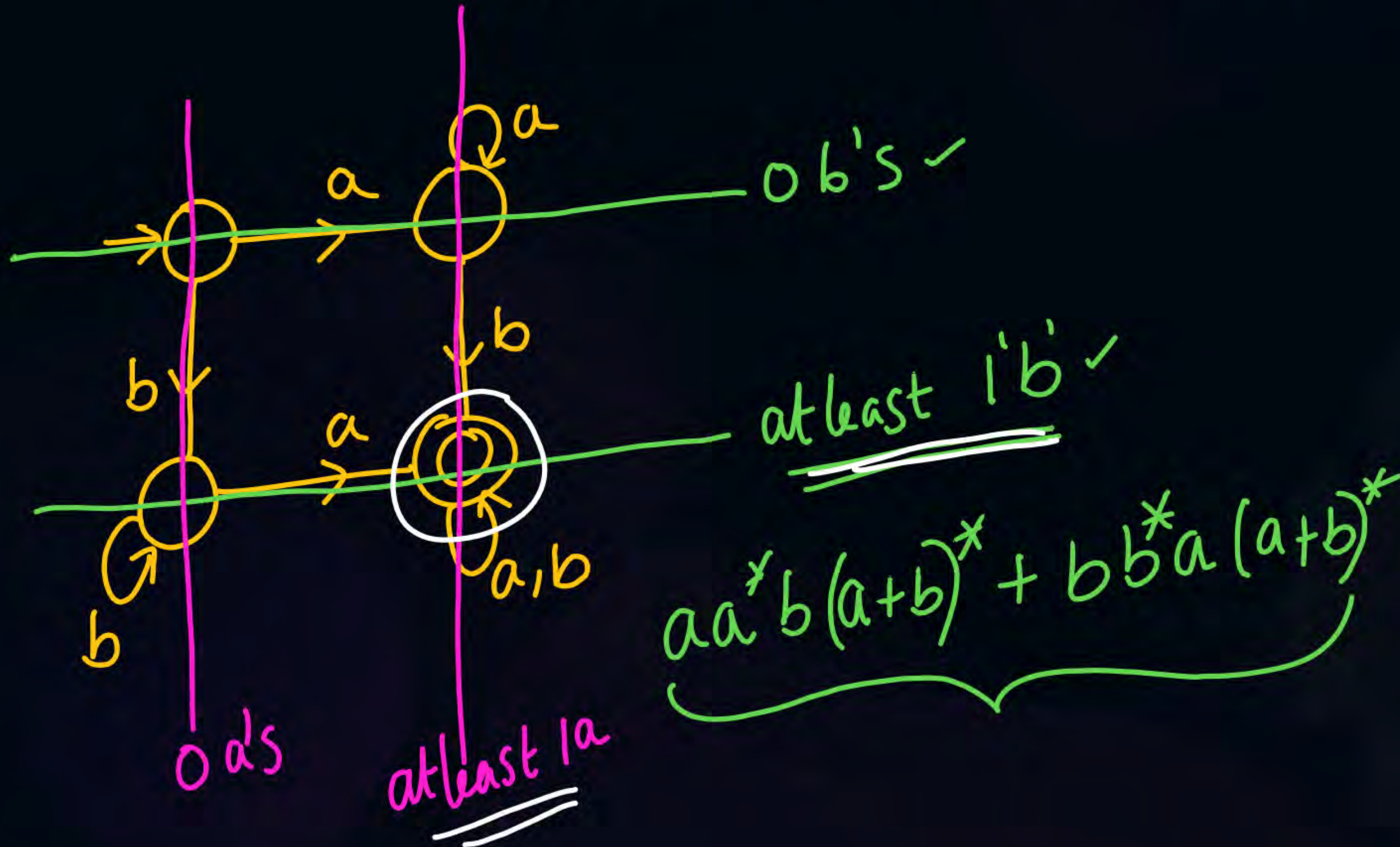
at least $\rightarrow$ no dead
 exactly and at most $\rightarrow$ dead

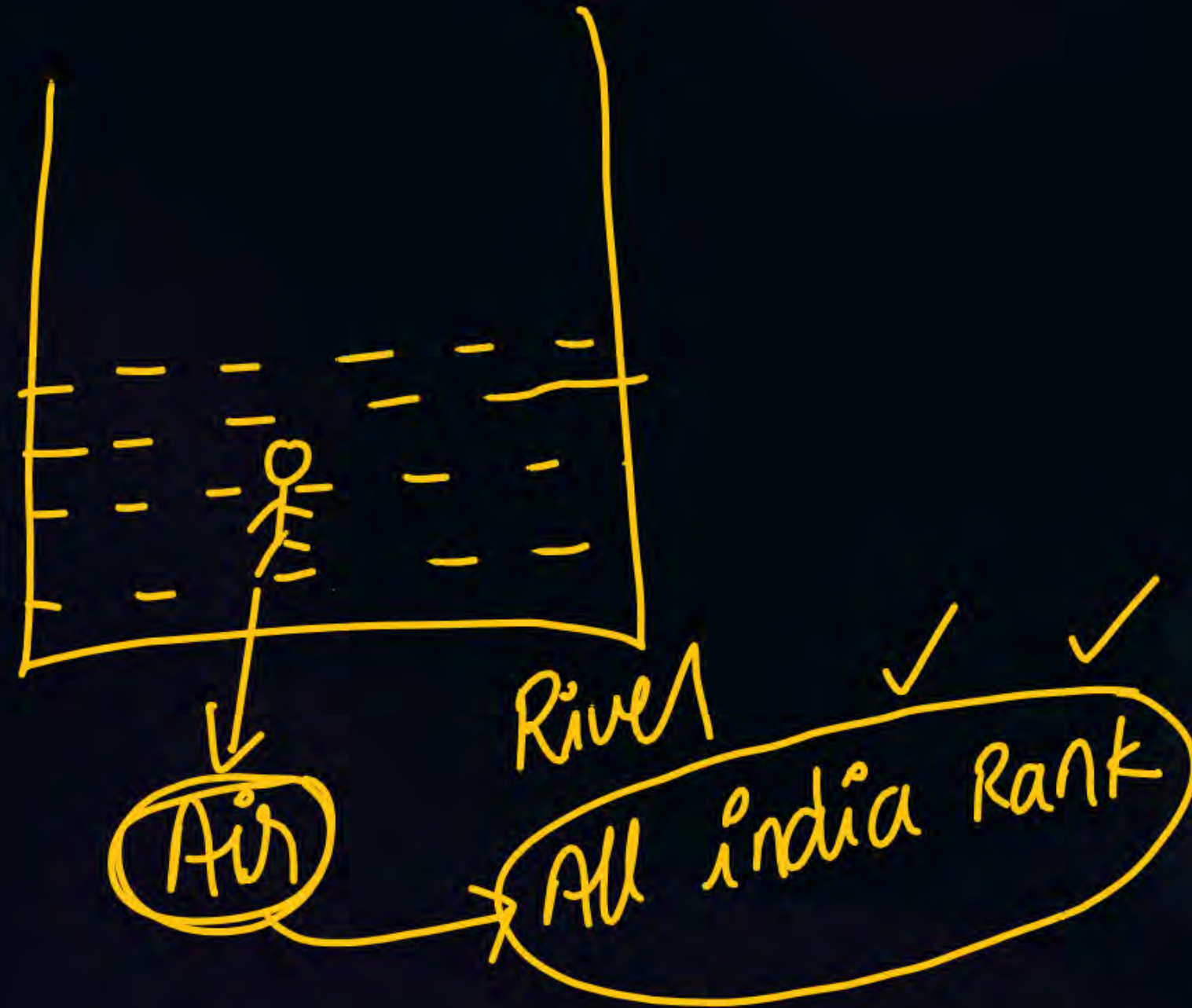


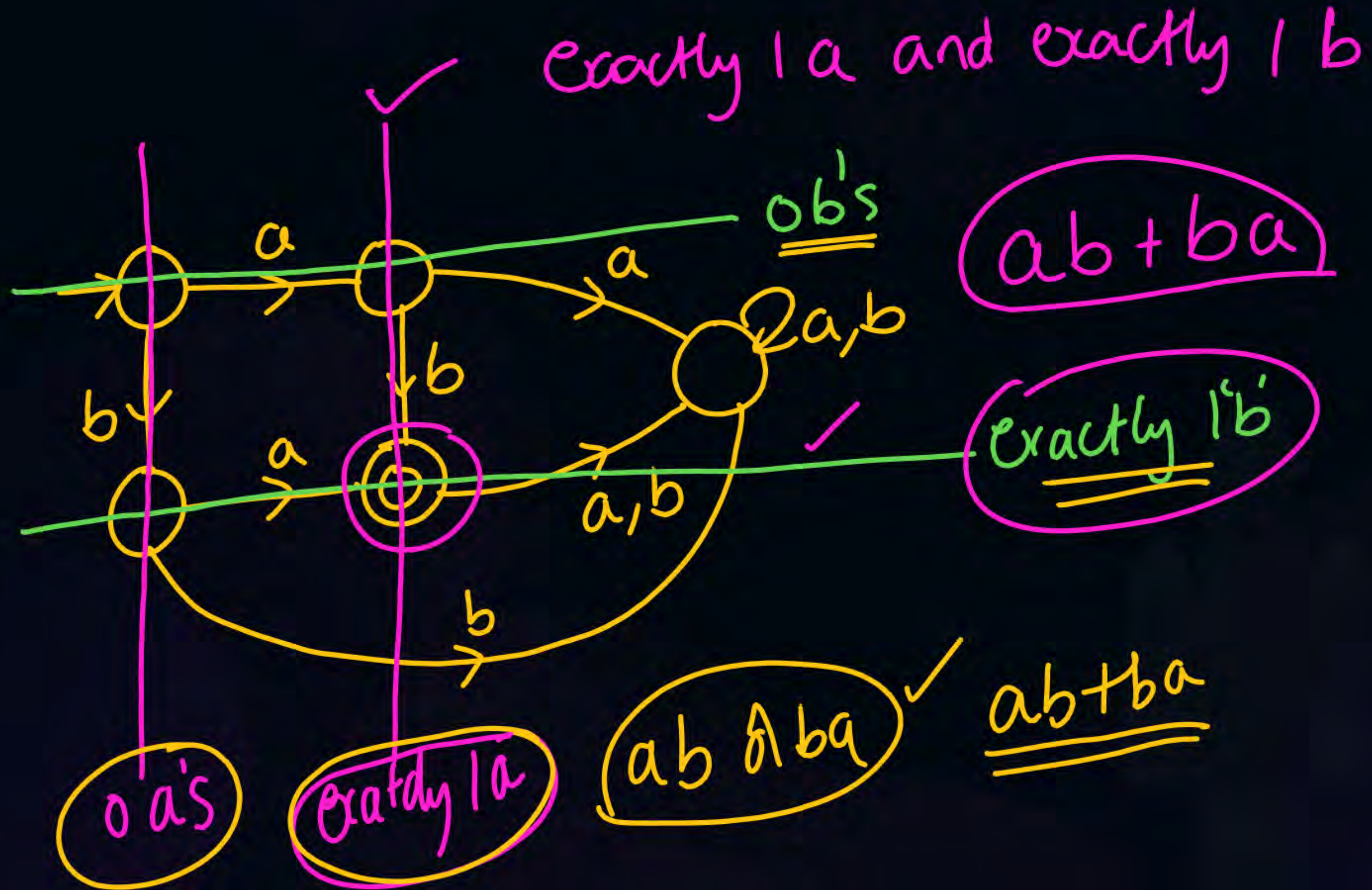
grid

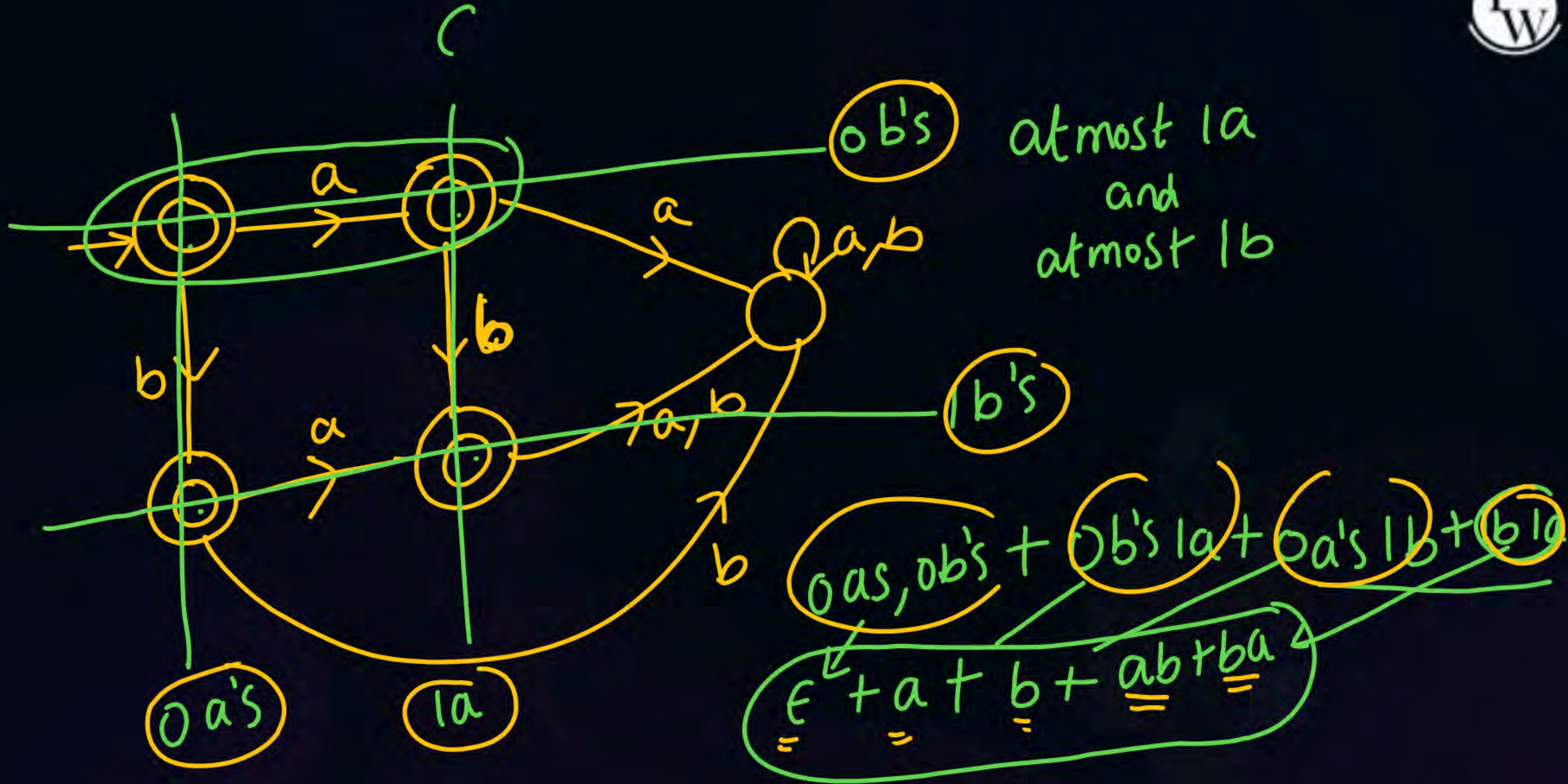
RE is not required for grid
RE is complex for grid

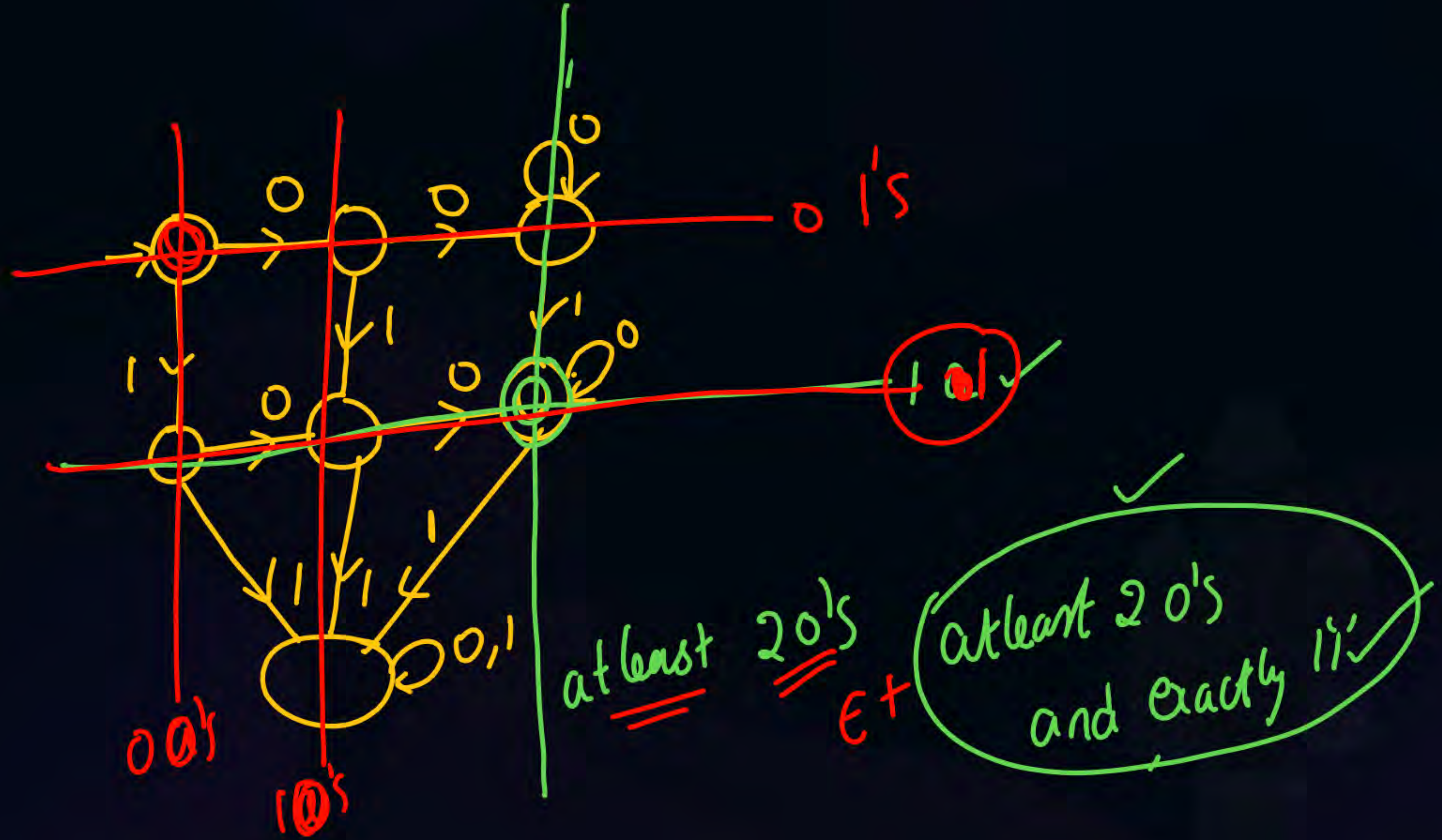
atleast 1a and atleast 1b

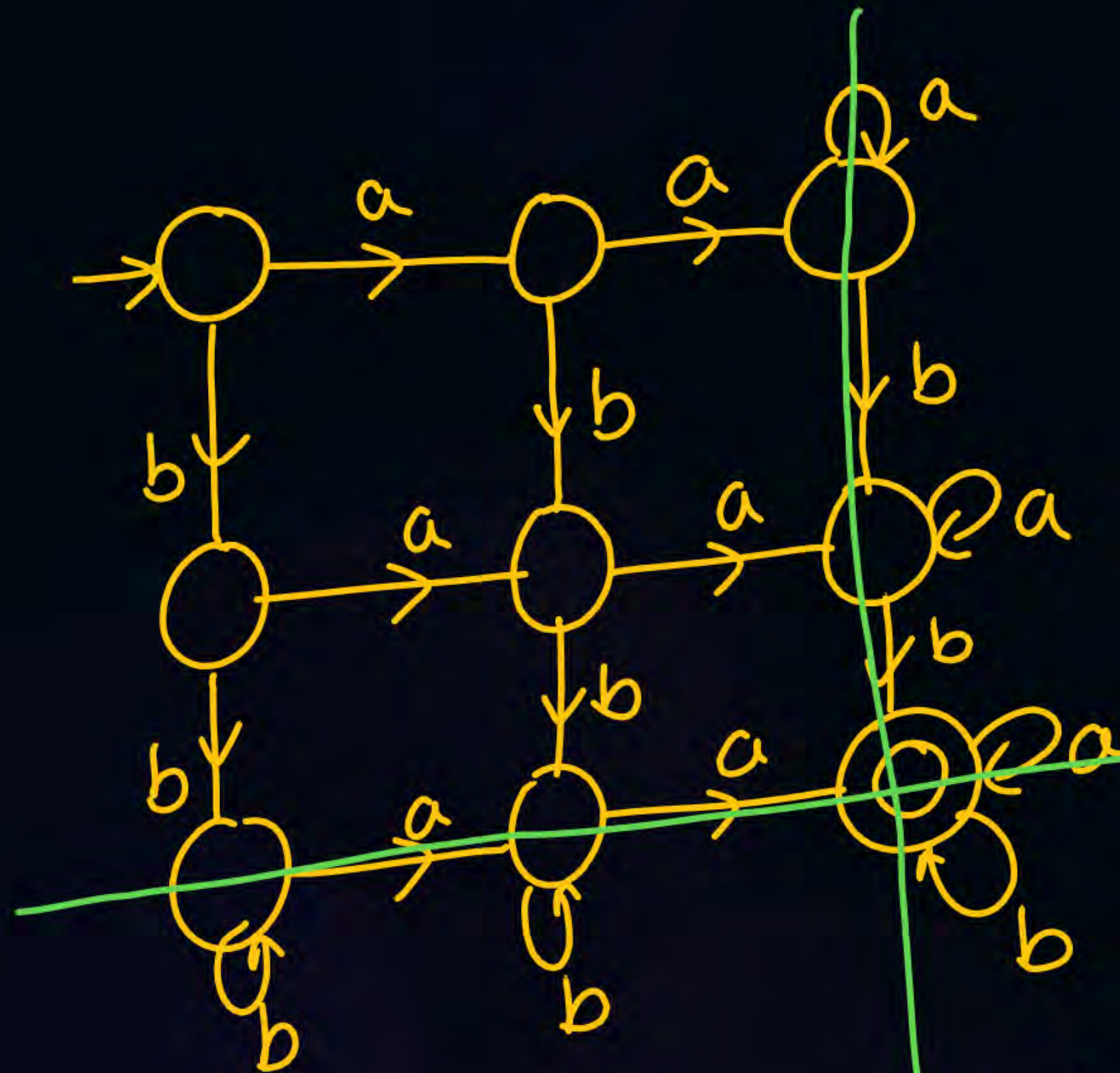








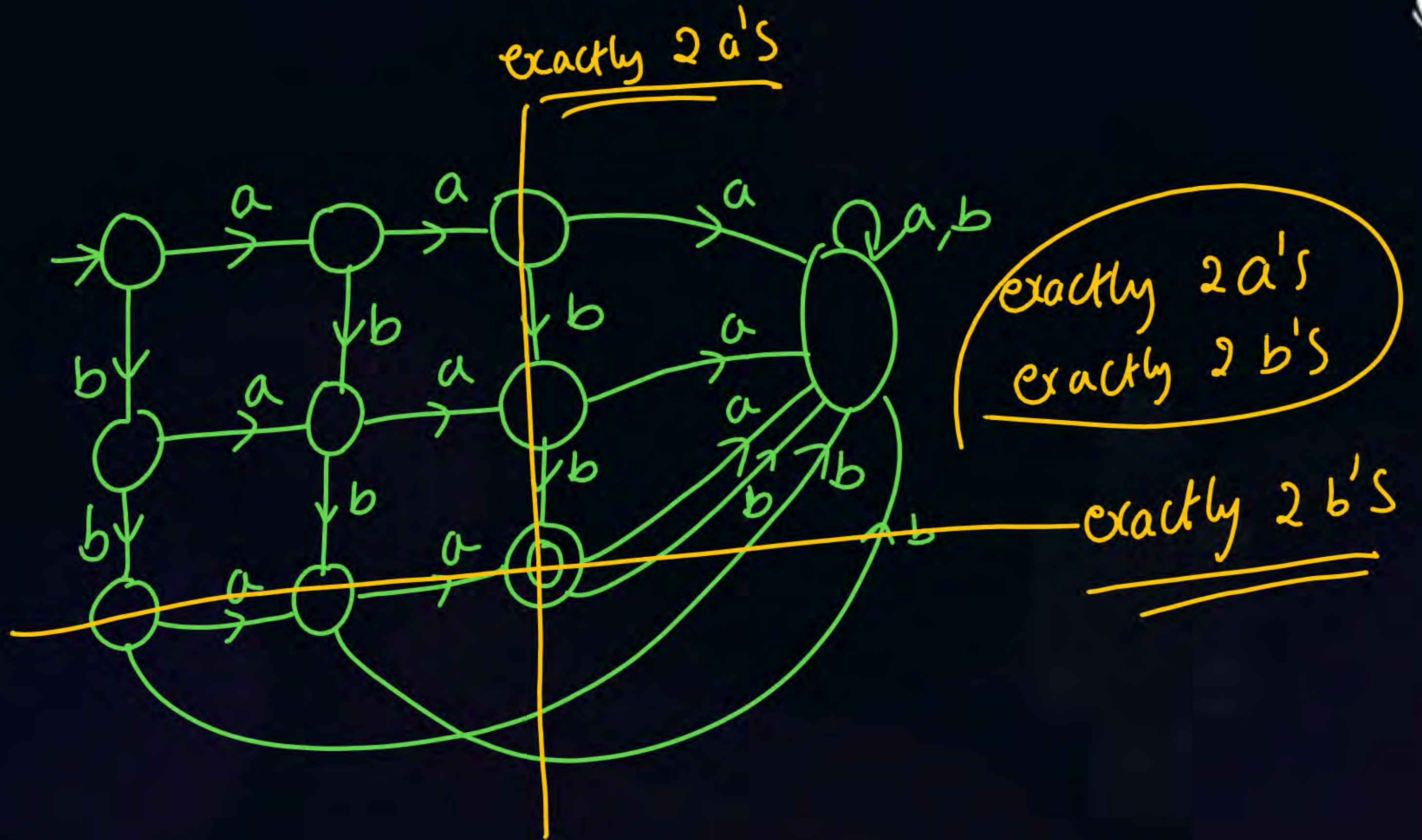




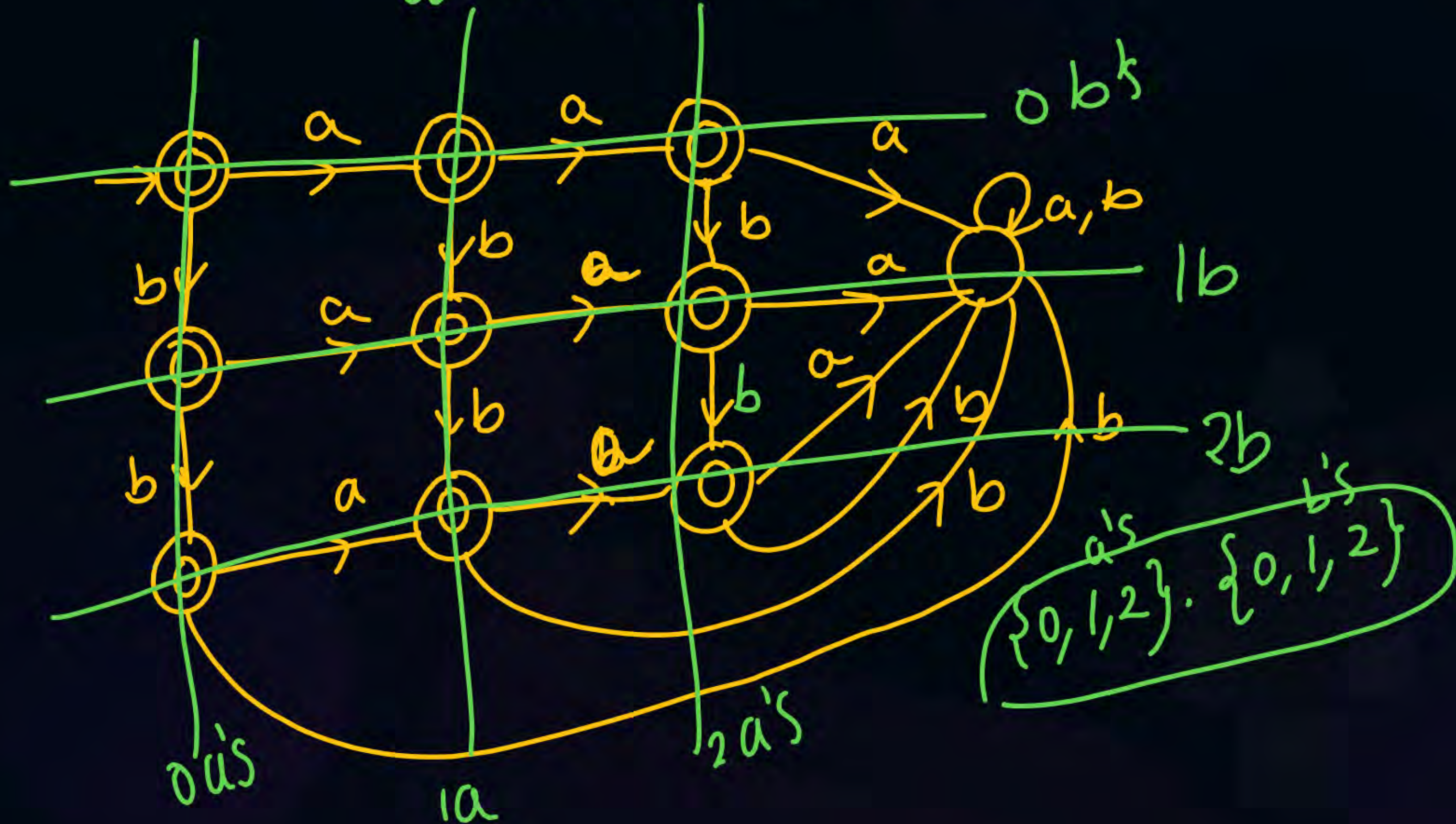
at least 2 a's at least 2 b's

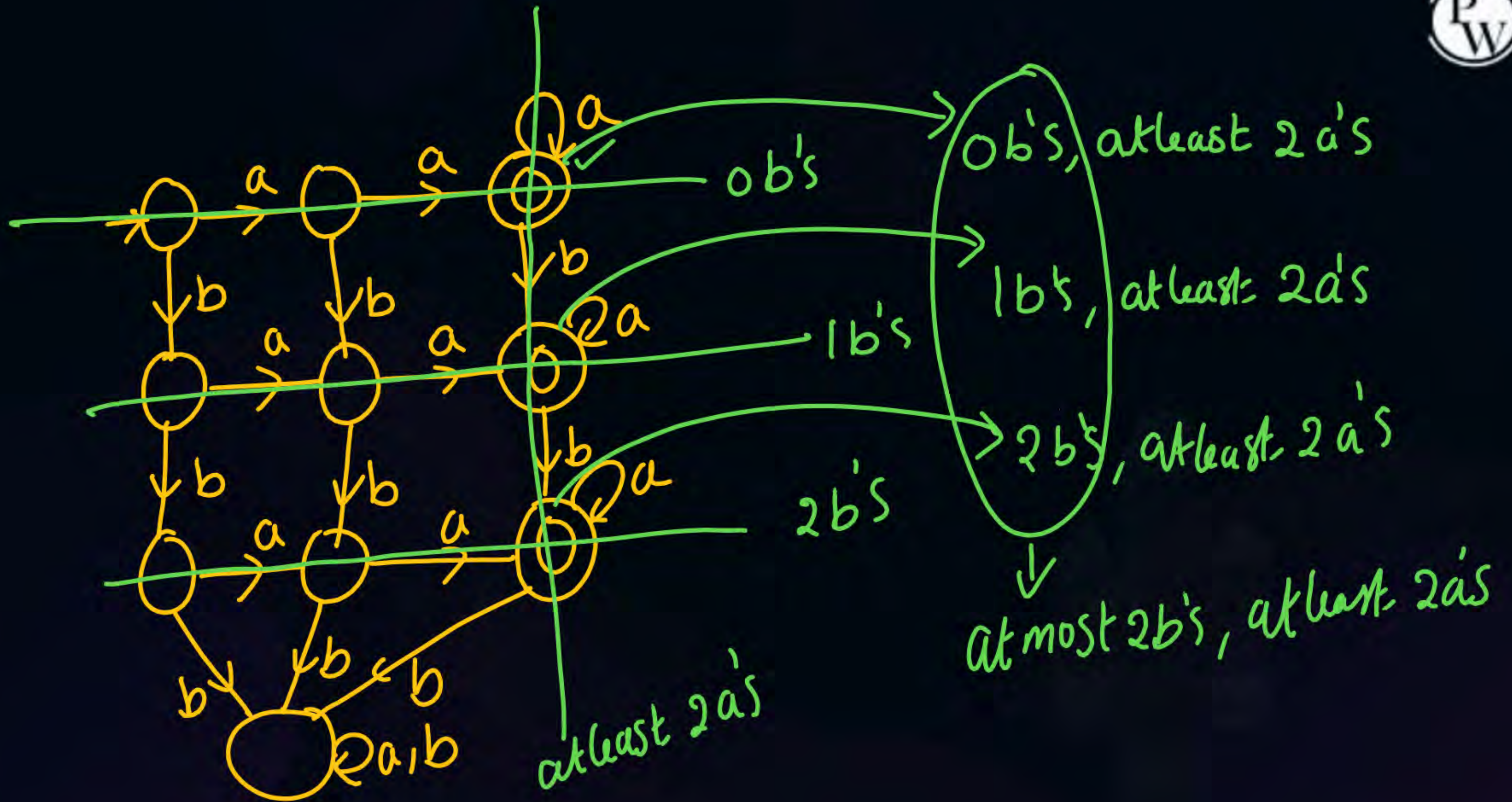
at least 2 b's

at least 2 a's

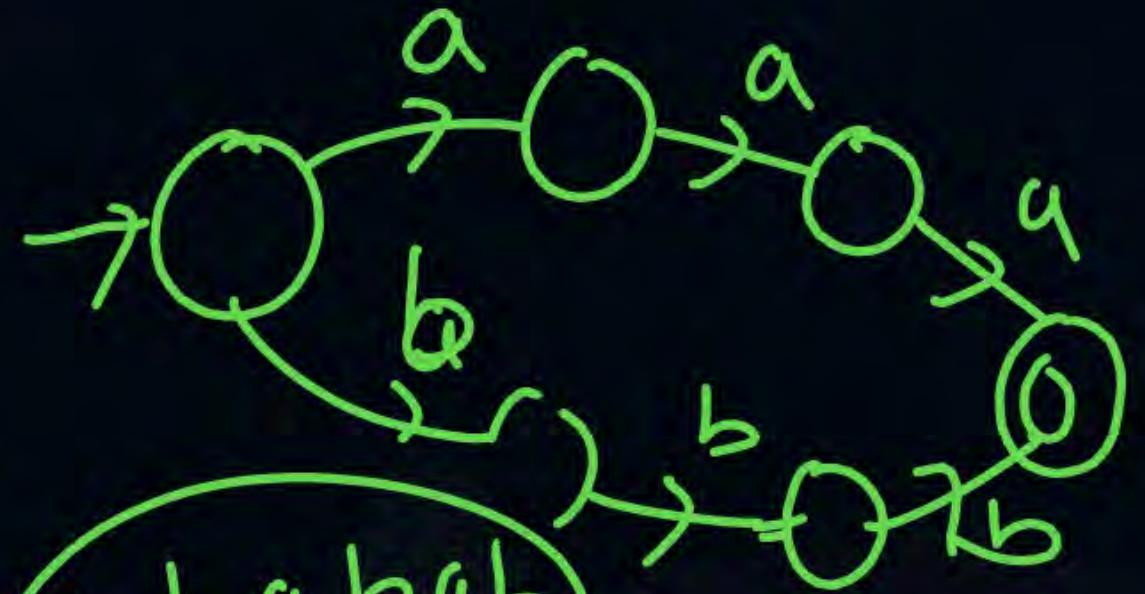
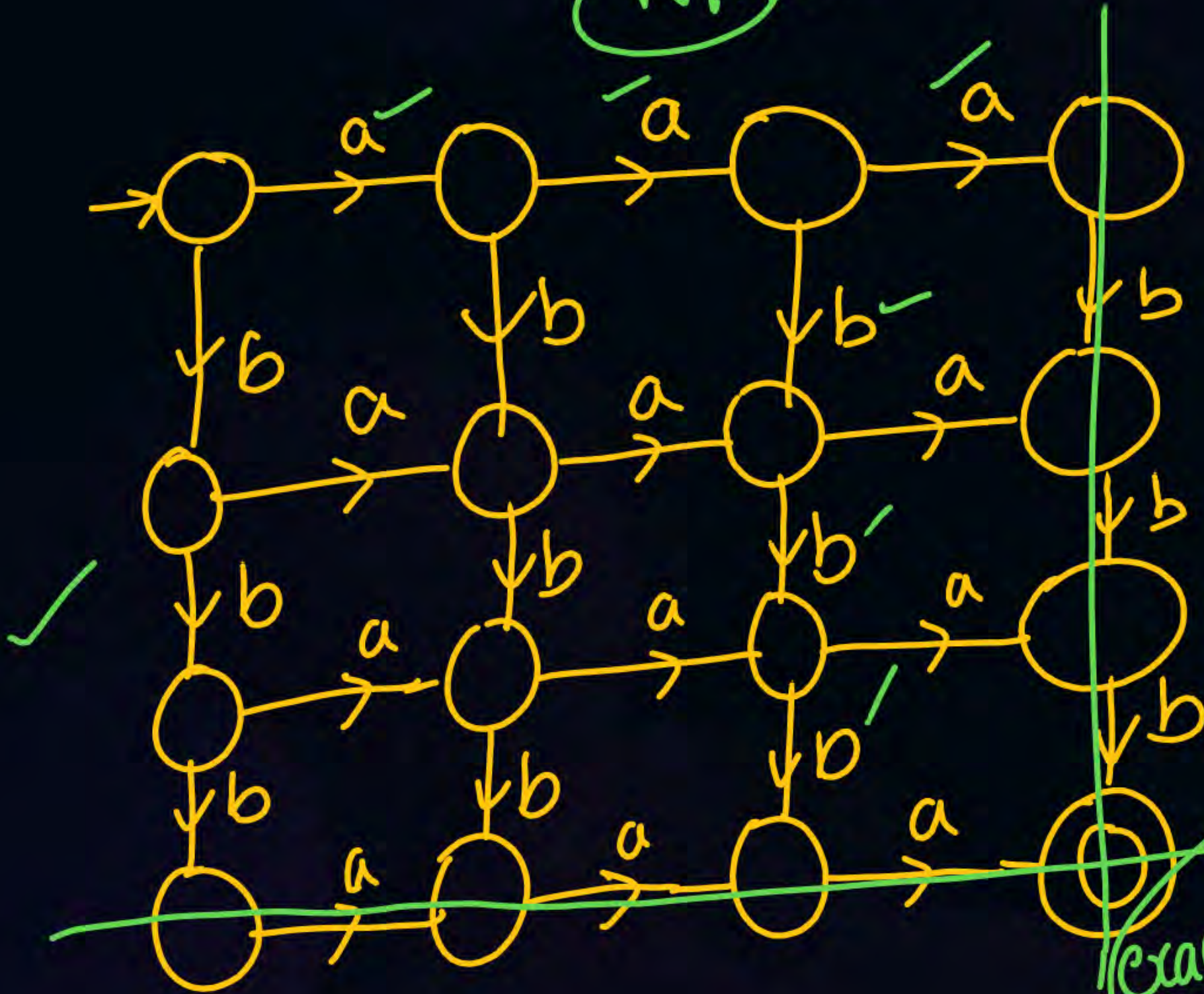


atmost 2a's atmost 2b's





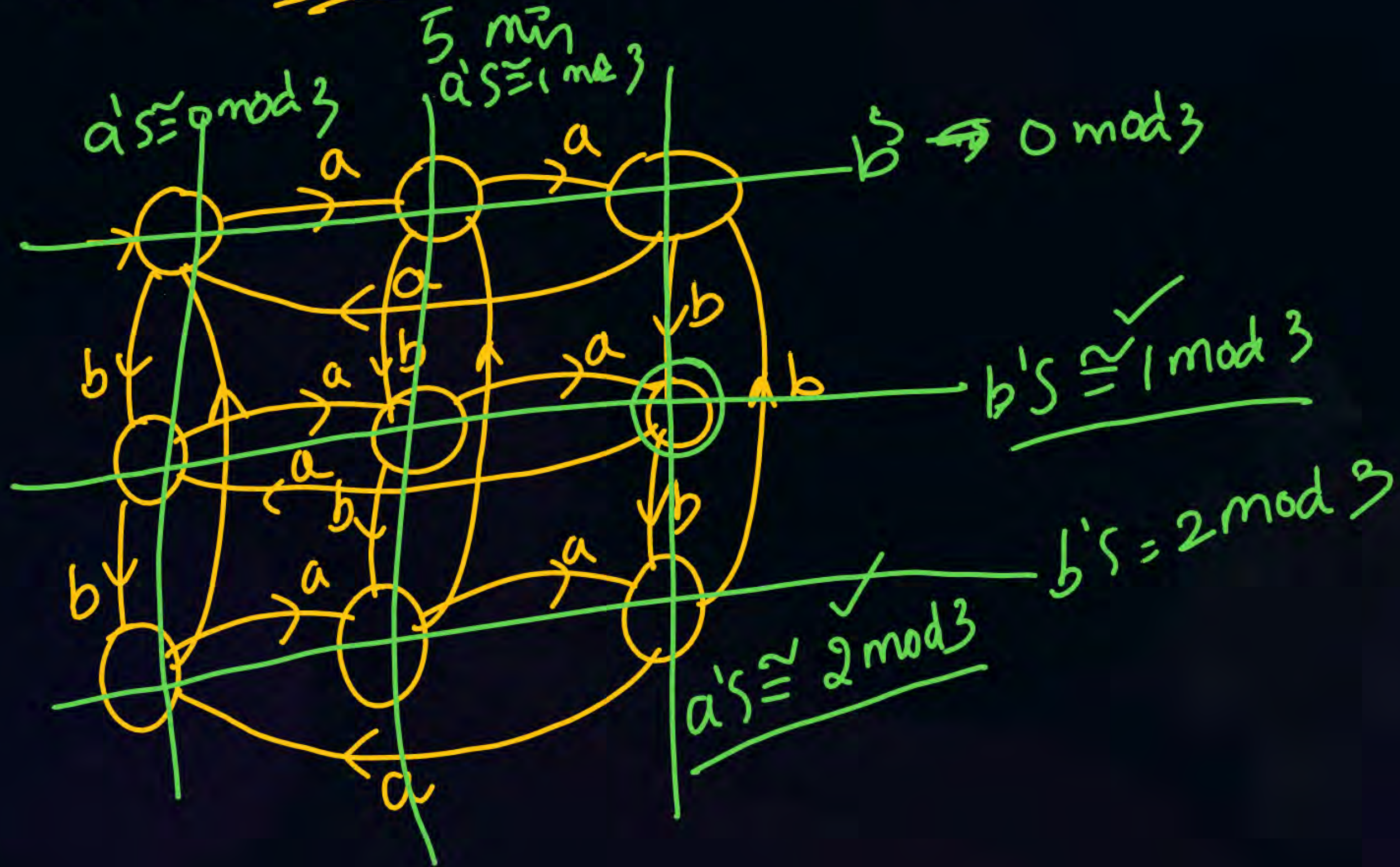
NFA



ababab

exactly 3 a's
exactly 3 b's

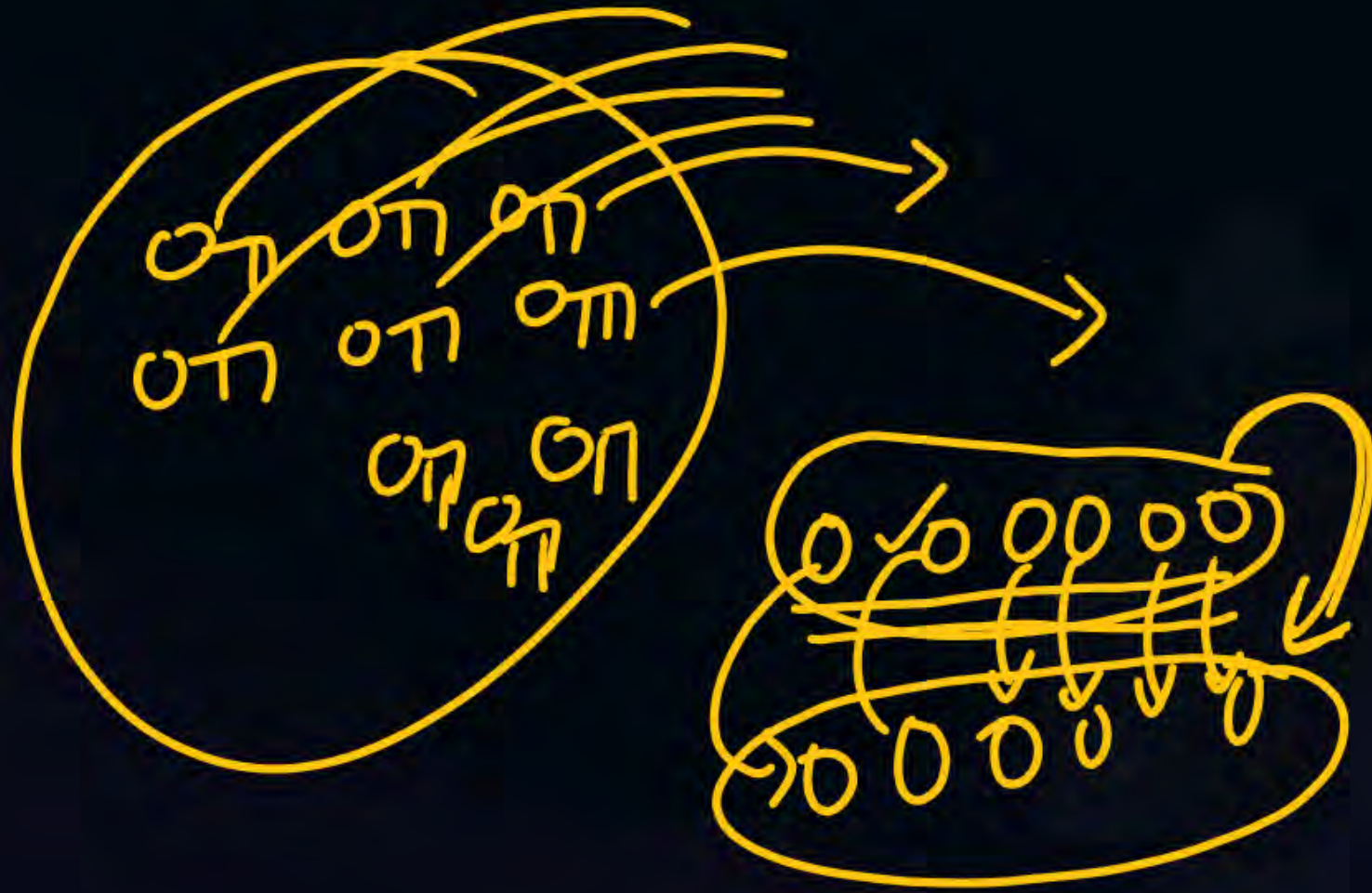
$$a's \equiv 2 \pmod{3} \quad b's \equiv 1 \pmod{3}$$



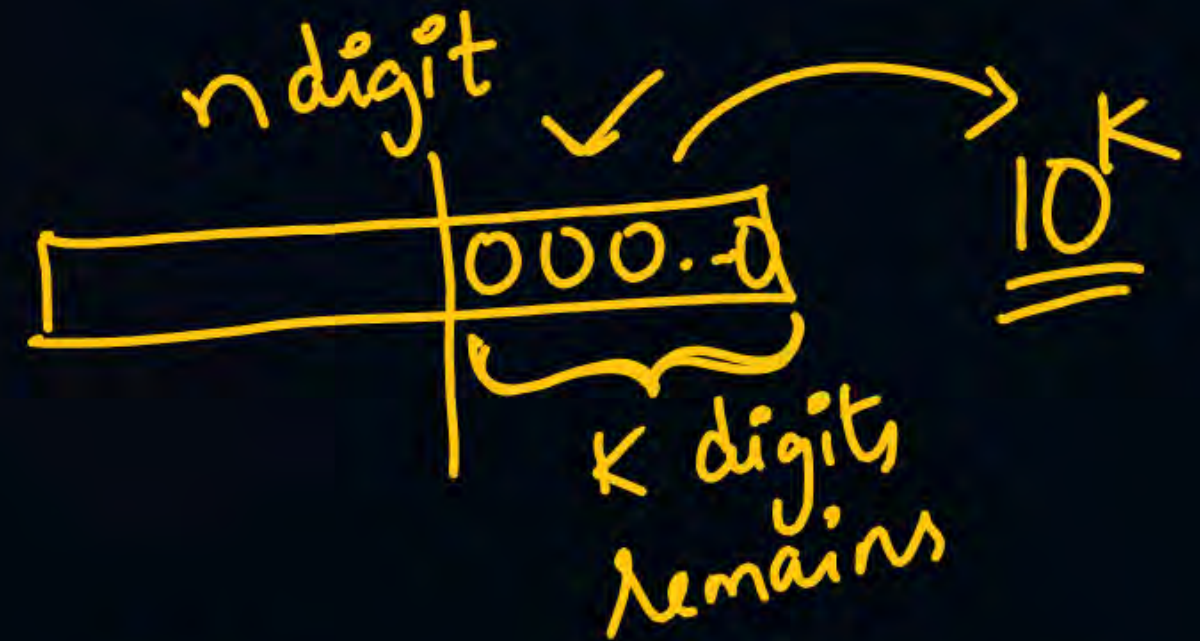
base 1 — 0000 - 4
 000 - 3
 00 - 2
 0 - 1

Base 1
 $\Sigma = \{0\}$ ✓

0 T T T T
 100 cows
 0 ✓ 0 0 0 0 0 0 ... 0



Base (10): - Fingers $\rightarrow 10$ ✓



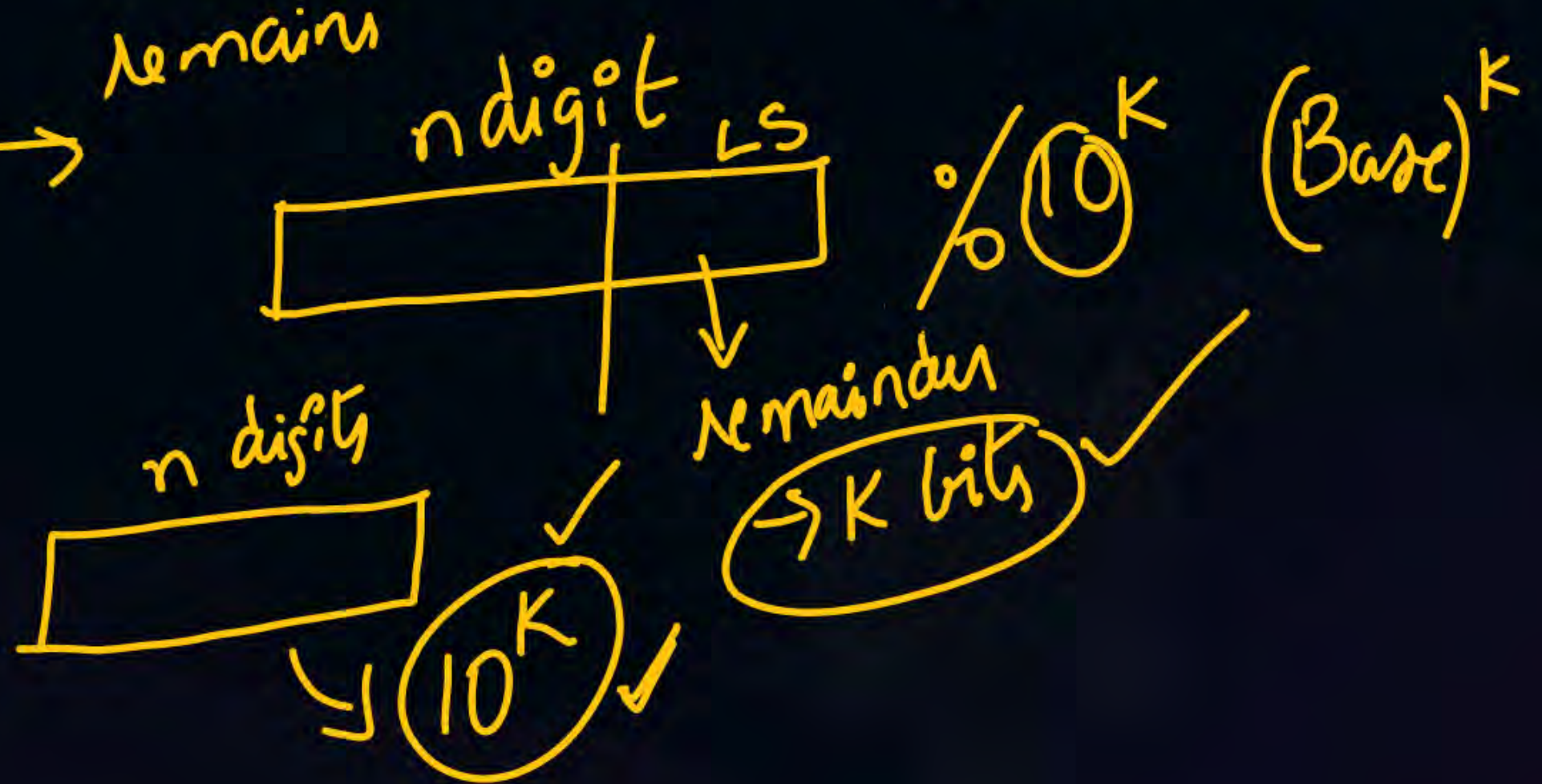
Base 10 $\rightarrow 0, 1, 2, 3, \dots, 9$

123 $\% 10 \rightarrow$ remains

123 $\% 10^2$

123 $\% 10^3$

123



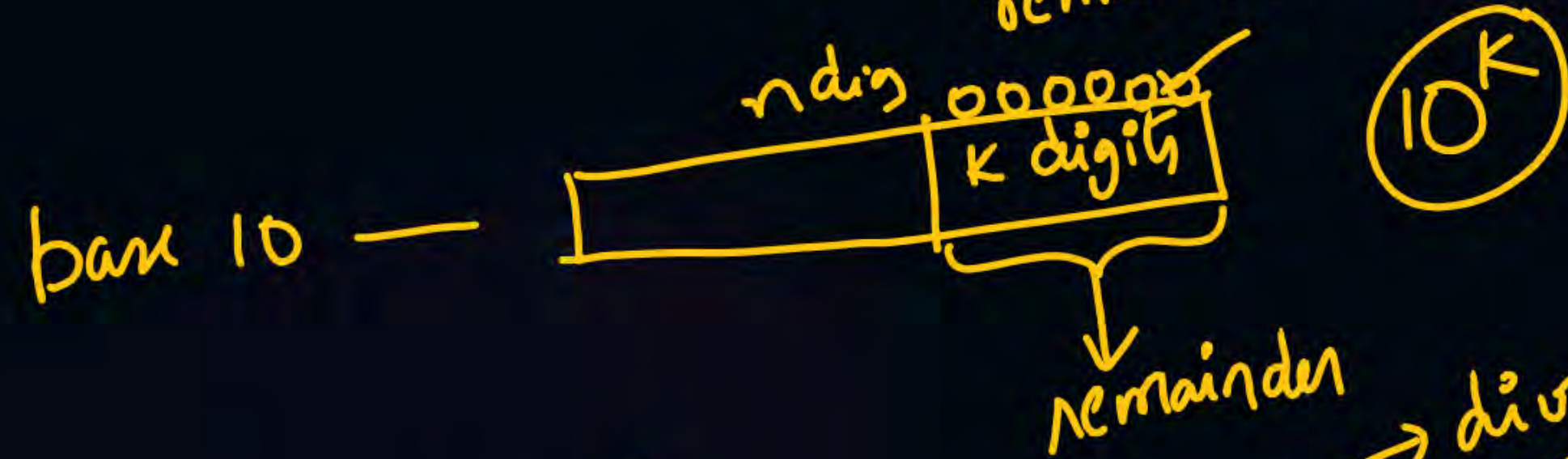
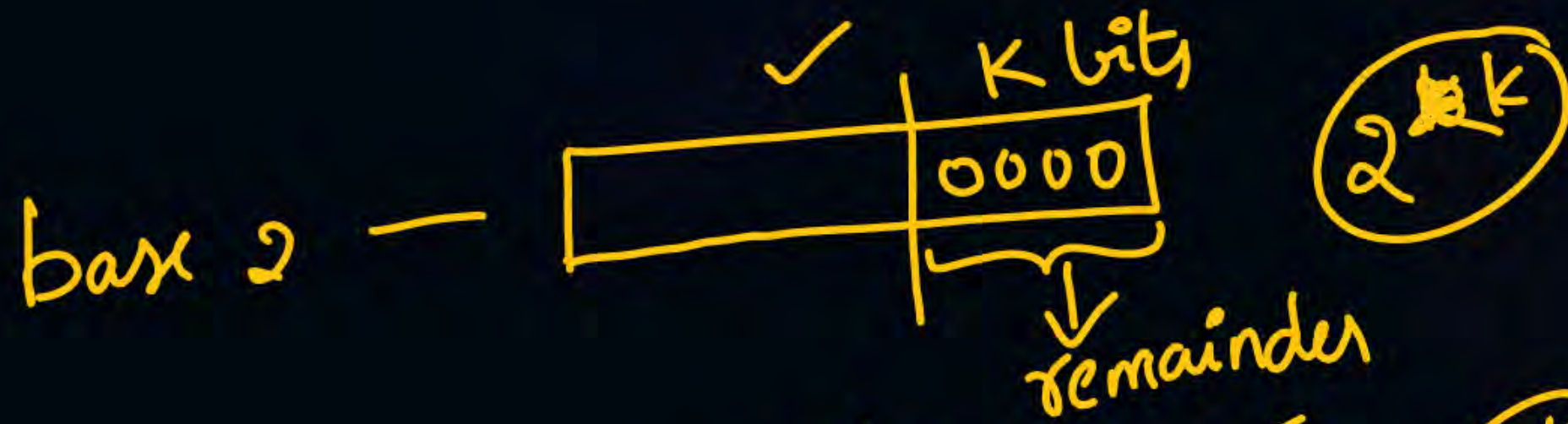
$$\Sigma = \{0, 1\}$$

$$\text{Base}(2) \rightarrow \{0, 1\} \rightarrow (2^k) \checkmark$$



↓
Remainder

$$\rightarrow (2^k) \% 0 \checkmark$$



Bank 2 $\rightarrow 2^k$ ✓

Bank 3 $\rightarrow 3^k$ ✓

Bank 10 $\rightarrow 10^k \rightarrow$ LS Kbits

$$\Sigma = \{\underline{0}, \underline{1}\} \rightarrow \text{Base } 2 \checkmark$$

$$L = \{ \text{Binary numbers divisible by } \textcircled{2} \}$$

$$2 \text{ LSB} \rightarrow 0$$



$$\text{Base } 2 \quad \textcircled{2^k}$$

How many states are in min DFA, that
accepts binary numbers divisible by 2^4

$(0+1)^* \underline{0000} \checkmark$

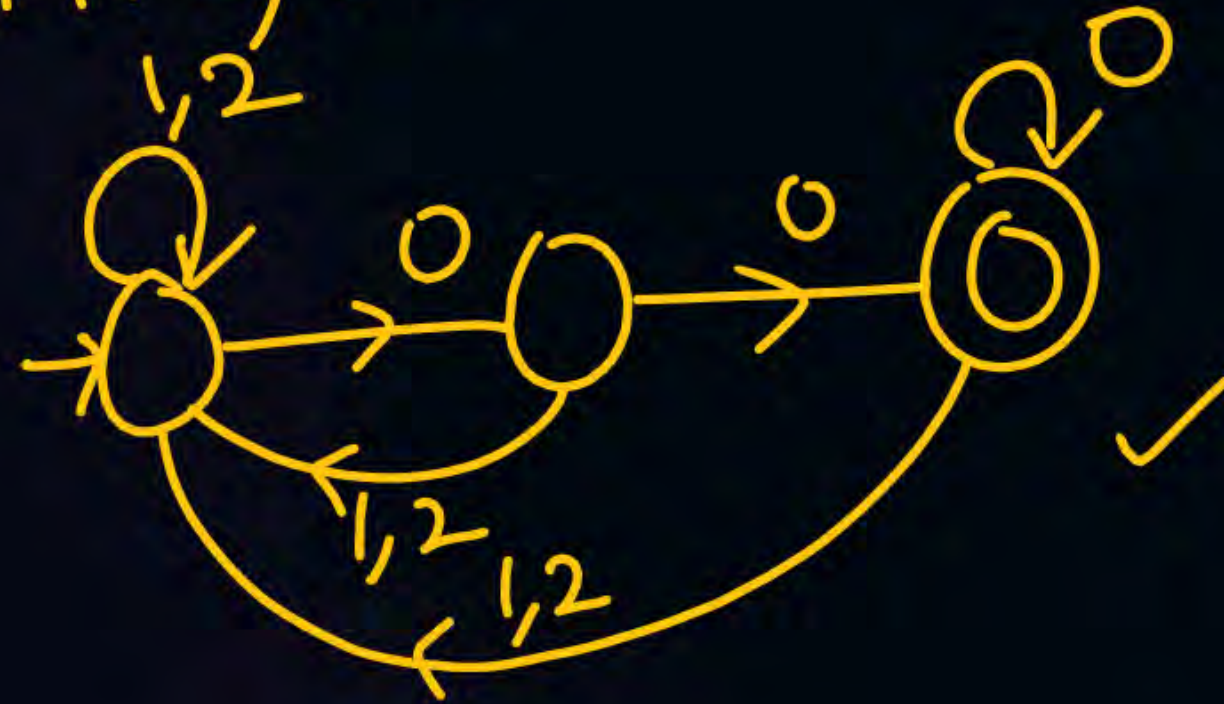


How many states are in min DFA
that accepts binary number divisible by 2^k ?

$(k + 1)$ states

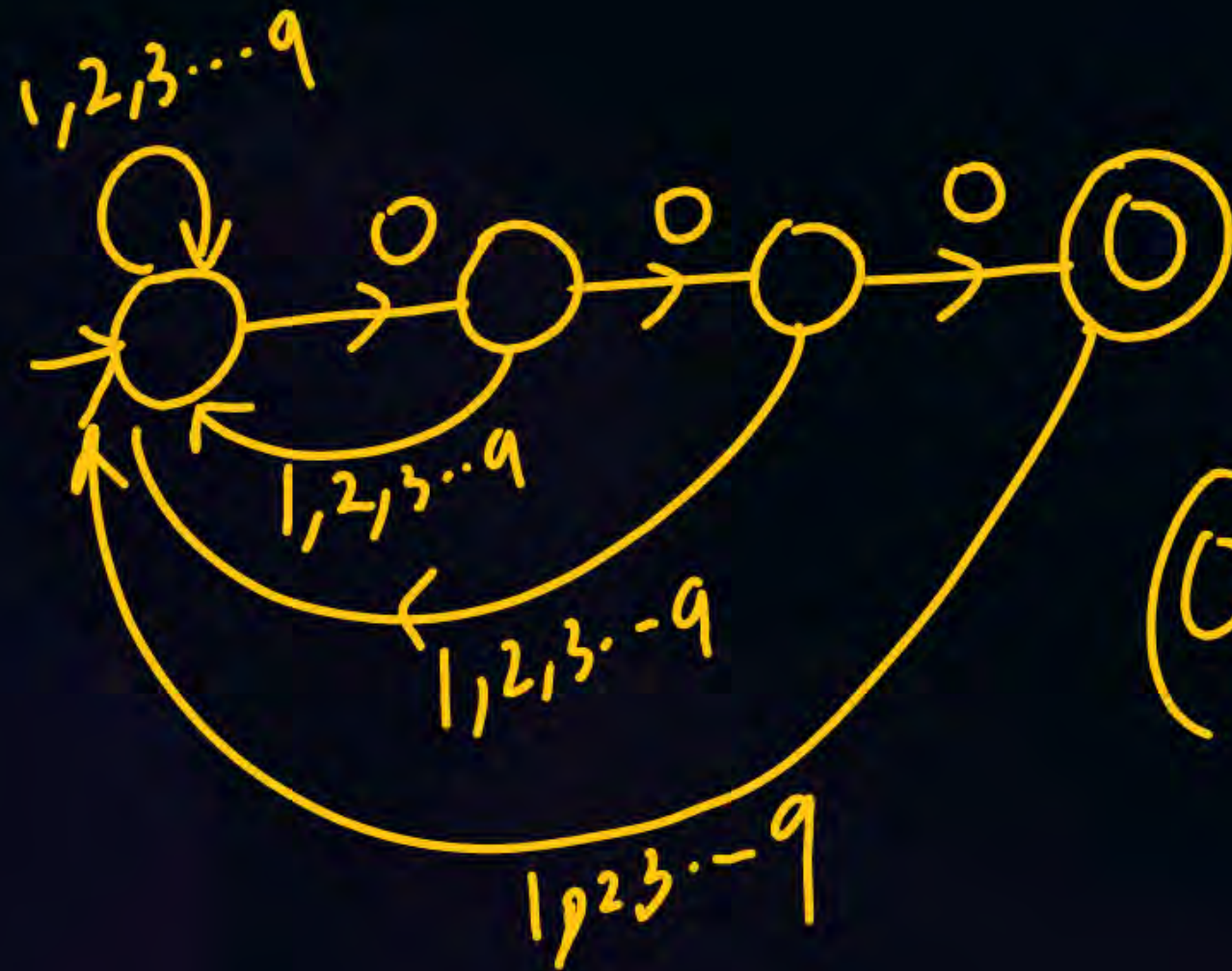
$\Sigma = \{0, 1, 2\}$ $\rightarrow$ ternary numbers $\rightarrow$ divisible by 3^2

$(0+1+2)^*00$



$\Sigma = \{0, 1, 2, 3, \dots, 9\} \rightarrow$ base 10 number system $\rightarrow$ divisible by 10^3

$\downarrow$
4 states



$(0 \neq 1 + \dots + 9)^* 000$

Base $n \rightarrow n^k \rightarrow$ last k digits '0'

$\Sigma = \{0,1\} \rightarrow$ binary language $\rightarrow$ divisible by 3.



$\Sigma = \{0, 1, 2, 3\} \rightarrow$ Base 4 number $\rightarrow$ divisible by '3' ✓

$\% K$

K -states

$\epsilon /$

1^+

q_s ✓

$\rightarrow q_s$

	0	1	2	3 ✓
q_0 *	q_0	q_1	q_2	q_0
q_1	q_1	q_2	q_0	q_1
q_2	q_2	q_0	q_1	q_2
q_s	q_0	q_1	q_2	q_s ✓

min DFA

$\Sigma = \{0, 1, 2, 3\} \rightarrow$ Base 4 $\rightarrow$ divisible by 5

	0	1	2	3
$\rightarrow^* q_0$	q_0	q_1	q_1	q_3
q_1	q_4	q_0	q_1	q_2
q_2	q_3	q_4	q_0	q_1
q_3	q_2	q_3	q_4	q_0
q_4	q_1	q_2	q_3	q_4

$\Sigma = \{0, 1\} \rightarrow \text{binary numbers} \equiv 1 \pmod{3}$

	0	1
$\rightarrow q_0$	<u>q_0</u>	<u>q_1</u>
$* q_1$	<u>q_2</u>	<u>q_0</u>
q_2	<u>q_1</u>	<u>q_2</u>



THANK - YOU